

7. GOVERNANCE & RELEASE GATE · STPA Hazard Analysis · Bumpless Human Override · Release Case ('LOOP-01' → 'REL-01')

1. SENSING
Photons / Voltages to DMA · MIPI/SPI

2. PERCEPTION
ViT Encoders · DINOv2 · 3D Tokens

3. MEMORY
Temporal Belief · $SE(3)$ Trees · JEPAs

5. PLANNING (System 1.5 · MPU/NPU)
Diffusion Policies · ACT Chunking (H -steps) · Jerk Continuity

4. REASONING (System 2 · MPU)
Vision-Language Models · 3D Goals · Leases (t_{expire})

THE PROPOSAL-PERMISSION PRIVILEGE BOUNDARY

6. REFLEX (System 1 · MCU)
1 kHz Real-Time Loop · Control Barrier Functions ($h(x) \geq 0$) · Stopping Distance $d_{\text{stop}} \leq d_{\text{clearance}} \cdot \text{Veto}(u_t)$

THE PHYSICAL WORLD ($W_t \rightarrow W_{t+1}$) · Kinetic Momentum · Joule Heat · Friction · Collision Dynamics

Endogenous Sensory Shift (O_{t+1})